

E-Commerce Sales Analytics & Power BI Dashboard

1. Project Overview

This project analyzes e-commerce sales data to identify important sales trends, product performance, customer purchasing patterns, order-status distribution, and payment-method preferences.

The project covers the complete Data Analytics workflow, including data preparation, exploratory data analysis (EDA), data visualization, Power BI dashboard development, business insights, and recommendations.

The final outcome is an interactive Power BI dashboard that helps understand business performance and supports data-driven decision-making.

2. Problem Statement

E-commerce businesses generate large amounts of transaction data, but raw data alone does not provide clear information about business performance.

The objective of this project is to analyze e-commerce transaction data and answer questions such as:

- Which product categories generate the highest sales?
 - Which products perform best?
 - How do sales change over time?
 - What is the distribution of order statuses?
 - Which payment methods are most commonly used?
 - What relationships exist between quantity sold and sales?
 - What actions can improve business performance?
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3. Dataset Description

The dataset contains e-commerce transaction-level information covering the year 2023.

Main columns

Column	Description
transaction_id	Unique transaction identifier
order_date	Date of the order
customer_id	Customer identifier
product_category	Product category

Column	Description
product_name	Name of the product
quantity	Quantity purchased
unit_price	Price per unit
discount_pct	Discount percentage
shipping_cost	Shipping cost
payment_method	Payment method
order_status	Current order status
gross_sales	Sales before discount
discount_amount	Discount amount
net_sales	Sales after discount

Note: The dataset used in this project is a simulated e-commerce dataset created for analytics practice and contains intentionally introduced missing values, duplicates, and invalid records for the data-cleaning stage.

4. Tools Used

- Python
- Pandas
- NumPy
- Matplotlib
- Jupyter Notebook / Google Colab
- Microsoft Power BI
- DAX
- Git
- GitHub

5. Data Cleaning Process

The raw dataset was inspected before performing analysis.

Cleaning steps

1. Checked the number of rows and columns.
2. Examined column names and data types.
3. Converted `order_date` into the appropriate date format.
4. Identified missing values.
5. Filled missing `shipping_cost` values using the median.
6. Replaced missing `payment_method` values with `Unknown`.
7. Handled missing `customer_id` values.

8. Removed records with missing order dates where required for time-based analysis.
 9. Removed duplicate transactions.
 10. Removed invalid records with:
 11. `quantity <= 0`
 12. `unit_price <= 0`
 13. Created calculated business fields:
 14. Gross Sales
 15. Discount Amount
 16. Net Sales
 17. Performed final validation to ensure the processed dataset was suitable for analysis.
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6. Exploratory Data Analysis (EDA)

The following analyses were performed:

- Dataset structure and data-type analysis
- Missing-value analysis
- Duplicate analysis
- Descriptive statistics
- Sales and order analysis
- Product-category analysis
- Top-product analysis
- Order-status analysis
- Payment-method analysis
- Monthly sales trends
- Correlation analysis
- Outlier detection
- Distribution analysis

The EDA helped identify the major patterns and relationships within the dataset.

7. Data Visualizations

Multiple visualizations were created to communicate the findings clearly.

Main visualizations

1. **Net Sales by Product Category**
2. **Monthly Net Sales Trend**
3. **Top 10 Products by Net Sales**
4. **Order Status Distribution**
5. **Orders by Payment Method**

6. Quantity vs Net Sales Relationship

These visualizations highlight category performance, sales trends, product rankings, order outcomes, customer payment preferences, and relationships between sales variables.

8. Power BI Dashboard

An interactive Power BI dashboard was developed to provide a business-focused view of the dataset.

KPI Cards

The dashboard includes:

- **Total Net Sales**
- **Total Orders**
- **Total Quantity Sold**
- **Average Order Value**

Dashboard Visualizations

- Monthly Net Sales Trend
- Net Sales by Product Category
- Top 10 Products by Net Sales
- Order Status Distribution
- Orders by Payment Method
- Quantity vs Net Sales Relationship

Interactive Slicers

The dashboard includes filters for:

- Product Category
- Order Status
- Order Date
- Payment Method

These slicers allow users to interactively explore different parts of the dataset.

9. Key Insights

Insight 1 — Electronics is the leading category

Electronics is the strongest-performing product category and contributes the largest share of overall net sales.

Insight 2 — High-value electronics products dominate sales

Products such as Smartphones, Monitors, Wireless Earbuds, Laptops, and Smartwatches are among the strongest contributors to sales.

Insight 3 — Monthly sales fluctuate

The monthly analysis shows noticeable variations in sales during the year, indicating that demand is not constant throughout the year.

Insight 4 — Most orders are successfully delivered

Delivered orders form the largest proportion of all orders, indicating generally good fulfillment performance.

Insight 5 — Cancellations and returns require attention

Cancelled and returned orders represent a meaningful portion of transactions and can negatively affect revenue and customer satisfaction.

Insight 6 — Digital payments are highly important

Credit Card and UPI are the most frequently used payment methods, demonstrating strong customer preference for digital payment options.

Insight 7 — Category performance is uneven

Electronics significantly outperforms other categories, while categories such as Sports and Beauty & Health contribute comparatively less revenue.

10. Business Recommendations

Recommendation 1 — Focus on high-performing products

Maintain sufficient inventory of high-performing Electronics products and use targeted promotions to increase sales while reducing the risk of stock-outs.

Recommendation 2 — Reduce cancellations and returns

Investigate the major reasons for cancelled and returned orders. Improvements in product information, product quality, delivery processes, and customer support can help reduce unsuccessful orders.

Recommendation 3 — Strengthen digital-payment promotions

Since UPI and Credit Card are widely used, targeted cashback, discounts, or loyalty benefits can be used to encourage digital payments and repeat purchases.

Recommendation 4 — Use seasonal demand for planning

Monthly sales patterns should be used to plan inventory, marketing campaigns, and promotional activities according to expected demand.

11. Conclusion

This project demonstrates a complete data analytics workflow from raw data preparation to business decision-making.

Python was used for data cleaning, EDA, statistical analysis, and visualization, while Power BI was used to create an interactive business dashboard.

The analysis shows that Electronics is the major revenue-driving category, digital payments are highly preferred, and there is an opportunity to improve business performance by managing high-performing inventory and reducing cancellations and returns.

Overall, the project demonstrates how data analytics and visualization can transform transaction data into meaningful business insights.

12. Project Structure

```
E-Commerce-Sales-Analytics/  
|  
├── data/  
│   ├── raw/  
│   │   └── ecommerce_raw_sales.csv  
│   └── processed/  
│       └── ecommerce_cleaned_sales.csv  
├── notebooks/  
│   ├── data_cleaning.ipynb  
│   └── EDA.ipynb  
├── visualizations/  
│   └── charts/  
├── powerbi/  
│   └── ecommerce_sales_dashboard.pbix  
├── presentation/  
│   └── project_presentation.pptx  
├── README.md  
└── requirements.txt
```

13. Skills Demonstrated

- Data Cleaning
- Exploratory Data Analysis
- Statistical Analysis

- Data Visualization
- Business Intelligence
- Power BI
- DAX
- Dashboard Development
- Business Insights
- Data-Driven Decision Making
- Git & GitHub